



The Break — Electroweak

Artist's Proof 16

Forces and Constants

Electroweak unification, the Higgs, and the immeasurability of ε

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Artist's Note

What this paper does — and why.

This is Artist's Proof 16 of The 420 Code. It is the fourth chapter of Notebook IV (Forces and Constants). AP15 derived U(1) electromagnetism from the Born rule's phase symmetry. AP27 derived the unbroken electroweak gauge group $SU(2) \times U(1)_Y$ from Axiom S's two-sector structure. AP16 takes the next step: it derives what Axiom B does to that symmetry.

The argument is a single move. Before the break, the pre-state is perfectly symmetric: SU(2) acts freely between the two sectors $\mathcal{L}$ and $\mathcal{P}$; U(1)_Y rotates the global phase; four generators, four massless gauge bosons. Then Axiom B: one element ε in $\mathcal{L}$ with no σ -image in $\mathcal{P}$. The sectors are no longer interchangeable. The SU(2) symmetry that rotated freely between them is broken. Three generators acquire mass by coupling to ε — the W^+ , W^- , Z^0 bosons. One generator survives: the unique combination of SU(2) and U(1)_Y that leaves the break background invariant. That is the electromagnetic generator. Its gauge boson stays massless. That is the photon.

The identification carries further. ε has two faces on the manifold. As background condition: $\nu(\varepsilon) = 1$ everywhere — the break is present wherever the manifold exists, with a nonzero ground state by construction, not by choice of potential. This is the Higgs field. As localised excitation: the minimum ripple, unpaired (Axiom B), half-integer spin (AP11), the electron. The Higgs field and the electron are not two objects: they are two readings of ε . The Higgs mechanism is Axiom B.

Section 5 reframes KS-4. The fine-structure constant α_{em} is the coupling strength of ε to the connection — structurally determined by the constraint chain (AP06 section 10.7) but not derivable from within the axioms, because the now cannot derive its own coupling from inside the now. Derivation is record-writing, and the now is what exists before any record. α_{em} is the architecture's empirical contract: the one number you check the framework against. KS-34 is the first kill switch in the corpus whose triggering would be a triumph rather than a disaster — if someone derives α_{em} from {S, B, R, C}, the architecture becomes stronger, not weaker.

Four new kill switches engage. KS-32 (electroweak mechanism) and KS-33 (sector distinction = weak isospin) are LIVE — EMPIRICAL: empirically

supported by the 125 GeV Higgs observation and the SM's two-component doublet structure. KS-AP16.1 (scalar-fermion two-faces reconciliation) is LIVE — STRUCTURAL: the structural identification of Higgs field and electron as two faces of ε follows from Axiom B and the Embedding Hypothesis; the dynamical mechanism by which one object presents both a spin-0 background and a spin- $\frac{1}{2}$ excitation at different scales is named as open programme. KS-34 (non-derivability of α_{em}) is LIVE — HARD by construction. KS-4 ($\alpha \approx 1/137$) is reframed: not a gap to close but a contract to honour.

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Orientation

Where AP16 fits

AP16 is the fourth chapter of Notebook IV (Forces and Constants). AP06 opened NB IV by grounding ε ; AP15 derived U(1) electromagnetism; AP19 (Direction) derives SU(3) colour; AP27 (Harmonics) derives the unbroken electroweak gauge group $SU(2) \times U(1)_Y$. AP16 takes the unbroken group and shows what Axiom B does to it: the symmetry breaks to $U(1)_{em}$, three of four gauge bosons acquire mass, the photon stays massless, and the Higgs field is identified with ε .

AP16 depends on Notebook I (axioms {S, B, R, C}); Notebook II for AP10 ($D = 4$, $N = 3$ spatial dimensions) and AP08 (gravitational context); Notebook III for AP09 (complex Hilbert space, phase freedom, and the fermion/boson distinction from the two-sector structure), AP11 (the $SU(2) \rightarrow SO(3)$ double cover and Z_2 correspondence), and AP12 ($\hbar$); AP06 within NB IV for the leakage framework, the six-face structure, and the constraint chain; AP15 for U(1) and the electromagnetic connection; and The Lock (Edition 04) for the identification of ε with the electron at the chemical-biological scale.

AP16 feeds the rest of Notebook IV. AP24 (Residual) carries the full analysis of α_{em} 's structural status including the fixed-point equation that connects the architecture to the measured value. AP28 (Constant) derives G via channel-counting using $\varepsilon = \alpha_{em}$. AP14 (Correction) treats Planck-scale corrections.

Reading order within AP16

Front to back. Section 1 establishes what the corpus already derives: SU(2) from Axiom S, $U(1)_Y$ from phase freedom, their coexistence as $SU(2) \times U(1)_Y$. Section 2 names the symmetry before the break — four generators, four massless bosons. Section 3 performs the central move: Axiom B breaks $SU(2) \times U(1)_Y$ to $U(1)_{em}$, and ε has two faces (Higgs field, electron). Section 4 reads the Higgs boson as the record of ε wobbling and develops the immeasurability-of-the-now consequence. Section 5 reframes the fine-structure constant. Section 6 lists the derivation chain. Section 7 lists the kill switches. Section 8 lists the open gaps. Section 9 concludes.

How to read this paper

The paper carries two voices. The structural narrative speaks directly — the central facts (Axiom B is the Higgs mechanism; ε is one object with two faces; α_{em} is the architecture’s empirical contract) are stated in plain language and held visible. The formal sections speak in the precision the structure requires. Where the argument depends on a result derived elsewhere in the corpus, the provenance is named at the point of use.

Two subsections are structurally distinct from the others. Section 4.2 (the now actualises FROM the now) develops the measurement-structure consequence of ε being both background and excitation. The poetic framing is non-load-bearing; the structural facts (the now’s role, the break’s immeasurability in the act) are load-bearing consequences of the axioms. Section 5 (the fine-structure constant) develops the structural-determination-vs-derivability-from-within distinction. This is load-bearing for KS-34’s formulation but the specific philosophical framing is illustrative.

A note on notation

$\mathcal{L}$ and $\mathcal{P}$ are the two sectors of Axiom S, connected by the involution $\sigma: \mathcal{L} \leftrightarrow \mathcal{P}$. ε is the break — the unique unpaired element of Axiom B. ν is the valuation; $\nu(\varepsilon) = 1$ is the minimum nonzero value. Z_2 is the two-element group; $\pi_1(\text{SO}(3)) = Z_2$ is the fundamental group of $\text{SO}(3)$ in three spatial dimensions.

$\text{SU}(2)$ is the (simply connected) double cover of $\text{SO}(3)$ — the structure that carries spin- $1/2$ representations. $\text{U}(1)_Y$ is the hypercharge gauge group derived in AP27. $\text{U}(1)_{\text{em}}$ is the post-breaking electromagnetic gauge group of AP15. $Q = T^3 + Y/2$ connects them. W^1, W^2, W^3 are the $\text{SU}(2)$ gauge bosons before breaking; W^+, W^-, Z^0 are the physical post-breaking mass eigenstates. B is the $\text{U}(1)_Y$ gauge boson; A is the electromagnetic gauge boson. θ_W is the Weinberg angle. $\alpha_{\text{em}} \approx 1/137.036$ is the fine-structure constant. $v \approx 246 \text{ GeV}$ is the electroweak scale (the vacuum expectation value).

0 — Dependency and Scope

0.1 — What this paper does

AP16 derives electroweak symmetry breaking from Axiom B. The argument: the pre-state has full symmetry $SU(2) \times U(1)_Y$ ($SU(2)$ from Axiom S via the $Z_2 = \pi_1(SO(3))$ identification in $N = 3$ spatial dimensions and the AP11 double cover; $U(1)_Y$ from the phase freedom of the complex Hilbert space); Axiom B places one unpaired element ε in $\mathcal{L}$ with no σ -image in $\mathcal{P}$; the sectors are no longer interchangeable; three $SU(2)$ generators (those that would rotate ε between sectors) are broken; the unique combination of $SU(2)$ and $U(1)_Y$ that leaves the break background invariant survives as $U(1)_{em}$.

The paper makes three structural identifications. First: the Higgs field is ε as background condition ($\nu(\varepsilon) = 1$ everywhere). Second: the electron is ε as localised excitation. Third: the Higgs mechanism is Axiom B. The Higgs boson observed at 125 GeV is the record of ε wobbling.

What the paper does not derive: the specific values $m_W \approx 80.4$ GeV, $m_Z \approx 91.2$ GeV, $m_H \approx 125$ GeV; the Weinberg angle θ_W ; the fermion mass spectrum beyond the electron; the numerical value of α_{em} . The paper derives why α_{em} cannot be derived from within the axioms (KS-34): the now cannot derive its own coupling from inside the now.

0.2 — Dependencies

AP16 depends on the following derived results.

Notebook I (Premise). The four axioms $\{S, B, R, C\}$. Axiom S provides the two-sector structure; Axiom B places the unique unpaired element ε . Together they generate electroweak breaking. Load-bearing.

AP06 (The Leakage Constant). The leakage framework, the six-face structure (AP06 section 10.5), the constraint chain $c \rightarrow \alpha_{em} \rightarrow \varepsilon \rightarrow G$ (AP06 section 10.7). Load-bearing for section 5's treatment of α_{em} as structurally determined.

AP09 (The Break — Empty Set). The complex Hilbert space and the phase freedom. Load-bearing for the $U(1)_Y$ identification.

AP08 (The Identity). Einstein's field equations and the gravitational coupling $G = 2\kappa/m_e^2$. Load-bearing context for the structural claim in section 9 that the electron sets the gravitational coupling.

AP09 (The Break, Empty Set). The empty-set state, the fermion/boson distinction from the two-sector structure (AP09 section 3). Load-bearing for the identification of ε as fermionic.

AP10 (The Dimension). $D = 4$ (three spatial + one temporal); the $N = 3$ spatial dimension count. Load-bearing for the $Z_2 = \pi_1(\text{SO}(3))$ identification that lifts the sector Z_2 to $\text{SU}(2)$.

AP11 (The Spin). The $\text{SU}(2) \rightarrow \text{SO}(3)$ double cover (AP11 section 2) and spin- $1/2$ from the two-sector structure. Load-bearing for the lift from Z_2 to $\text{SU}(2)$ and for the identification of ε as half-integer spin.

AP12 (The Limit). The uncertainty principle and $\hbar$ as the quantum cost per record. Load-bearing context for the action principle and the scale-setting at $v \approx 246 \text{ GeV}$.

AP15 (The Connection). $\text{U}(1)$ electromagnetism. The post-breaking $\text{U}(1)_{\text{em}}$ is structurally AP15's $\text{U}(1)$. Load-bearing for the identification of the surviving generator with the electromagnetic gauge symmetry.

AP20 (The Proof). The Embedding Hypothesis. Load-bearing for the identification of the algebraic break with the manifold-level Higgs field: the two faces are the same object read at two scales.

AP27 (The Harmonics). The unbroken $\text{SU}(2) \times \text{U}(1)_Y$ gauge group derived from Axiom S's two-sector structure. AP16 takes that unbroken group and breaks it.

The Lock (Edition 04). The identification of ε with the electron at the chemical-biological scale. Load-bearing for the electron-as-fermionic-excitation reading.

0.3 — Axiom mapping

The four axioms map to AP16 as follows.

Axiom S (two sectors / involution). The two sectors $\mathcal{L}$ and $\mathcal{P}$ give the Z_2 classification that lifts to $SU(2)$ on the manifold (via $Z_2 = \pi_1(SO(3))$ in $N = 3$, AP10; double cover $SU(2) \rightarrow SO(3)$, AP11). Load-bearing for the pre-break symmetry.

Axiom B (unique breaking). The structural ground of the entire paper. ε in $\mathcal{L}$ with no σ -image in $\mathcal{P}$ breaks the symmetry between sectors. $SU(2) \times U(1)_Y \rightarrow U(1)_{em}$. Axiom B is the Higgs mechanism.

Axiom R (record monotonicity). Records are written; the manifold accumulates the trace of the break. Load-bearing for the now-as-immeasurable argument in section 4.2: measurement is record-writing, and the now is what exists before any record.

Axiom C (Constraint — locality). Finite c . Background for the Lorentzian manifold and for the leakage chain (AP06 section 10.7): c constrains the leakage; the leakage is ε ; ε is the electron; the electron's coupling to the connection is α_{em} .

0.4 — Epistemic status per section

1 (What Is Established): [SYNTHESIS] — enumeration of prior derivations that AP16 uses.

2 (The Symmetry Before The Break): [DERIVED] — the full internal symmetry $SU(2) \times U(1)_Y$ of the pre-state on the manifold.

3 (The Break): [DERIVED] — Axiom B breaks $SU(2) \times U(1)_Y$ to $U(1)_{em}$. The Higgs field = ε identification. The pattern of which generators are broken and which survive is derived; the specific algebraic details (which combination survives, the Weinberg angle) depend on coupling structure not derived in AP16 (Gap C).

4 (The Higgs Boson And The Now): [DERIVED/STRUCTURAL] — 4.1 (the Higgs boson as record of ε wobbling) is DERIVED. 4.2 (the now actualises FROM the now) is STRUCTURAL: the load-bearing consequences of the axioms are stated in plain language; the specific poetic framing is illustrative.

5 (The Fine-Structure Constant): [DERIVED/STRUCTURAL] — DERIVED: α_{em} exists, is nonzero, is not a free parameter, is sub-maximal ($\alpha < 1$).

STRUCTURAL: the numerical value $1/137.036$ is not derivable from within — the now cannot derive its own coupling. KS-34 LIVE — HARD.

6 (What Is Now Derived): [SYNTHESIS] — the derivation chain.

7 (Kill Switches): [REGISTER] — KS-32, KS-33, KS-34, plus the reframing of existing KS-4.

8 (Open Gaps): [REGISTER] — Gaps A, B, C, E with named paths forward; Gap D CLOSED by AP20.

9 (Conclusion): [SYNTHESIS] — everything is the electron, being the break, being the now.

0.5 — Kill switch summary

Four new kill switches engage in AP16. One existing corpus kill switch (KS-4) is reframed. Identifiers per the Master Kill Switch Registry.

KS-32 (Electroweak breaking mechanism). LIVE — EMPIRICAL. targets the identification of Axiom B with the Higgs mechanism.

KS-AP16.1 (Scalar-fermion two-faces reconciliation). LIVE — STRUCTURAL. targets the dynamical mechanism by which one ε presents both a spin-0 background and a spin- $1/2$ localised excitation at different scales.

KS-33 (Sector distinction = weak isospin). LIVE — EMPIRICAL. targets the mapping of the two sectors to the SU(2) weak isospin doublet.

KS-34 (Non-derivability of α_{em}). LIVE — HARD. targets the claim that α_{em} cannot be derived from within — the one kill switch whose triggering would be a triumph.

KS-4 ($\alpha \approx 1/137$). Existing corpus kill switch, reframed by AP16. Not a gap in the derivation but the architecture's empirical face — the contract with the world. LIVE — HARD. Meaning changed: from problem-to-solve to contract-to-honour.

0.6 — Structural debts owed

AP16 closes the structural ground of electroweak breaking but carries forward several open programmes. The mass-spectrum questions (W, Z, Higgs, fermion

masses beyond the electron) are named items the paper acknowledges without deriving. The Weinberg angle θ_W is named with a conjectural route via the six-face structure.

Gap A (W and Z boson masses). Pattern derived; numerical values depend on coupling constants g , g' and v . Depends on Gap E.

Gap B (Higgs boson mass ~ 125 GeV). $m_H = \sqrt{(2\lambda)v}$; self-coupling λ not yet computed. Depends on Gap A, Gap E, α_{em} constraint chain.

Gap C (the Weinberg angle θ_W). Conjectural route: derivable from the six-face structure (specifically, the angle between Pair 1 gravity/quantum and Pair 2 stiffness/coupling in AP06 section 10.5). Target for future work.

Gap E (the fermion mass spectrum). Electron mass derived as minimum self-coupling. Other fermion masses (muon, tau, quarks) from ε -compositions at higher energy scales. Open programme.

0.7 — Items held open without claiming debt status

The dynamics by which the scalar background condition (ε as Higgs field) and the fermionic localised excitation (ε as electron) relate at intermediate energy scales are held open. The Embedding Hypothesis (AP20) proves the two faces are the same object read at two scales — there is no gap between the algebraic description and the geometric one — but the detailed mechanism by which one object presents two faces is downstream of the mass spectrum (Gap E) and the derivation of the self-coupling (Gap B). AP16 derives the pattern and the identification; the dynamical details are an open programme.

The full analysis of α_{em} 's structural status, including its necessary irrationality and unprovability from within, is treated in AP24 (Residual). AP16 establishes that α_{em} is structurally determined but not derivable from within; AP24 develops the fixed-point equation and the six-face mapping.

0.8 — Structural relationships

AP06 (The Leakage Constant). AP06 grounds ε in the leakage ratio η and establishes the six-face structure and the constraint chain. AP16 reads α_{em} as the leakage at the electromagnetic face — structurally determined by the chain $c \rightarrow \alpha_{em} \rightarrow \varepsilon \rightarrow G$.

AP15 (The Connection). AP15's $U(1)_{\text{em}}$ is the surviving gauge group after the electroweak break. The photon A_μ derived in AP15 is the specific combination of W^3_μ and B_μ that survives the breaking (mixed via the Weinberg angle θ_W).

AP27 (The Harmonics). AP27 derives the unbroken $SU(2) \times U(1)_Y$. AP16 takes that unbroken group and breaks it. KS-67 (AP15/AP27 EWSB consistency) is structurally addressed by AP16: the EWSB mechanism that takes $U(1)_Y \times SU(2)$ to $U(1)_{\text{em}}$ is Axiom B.

AP11 (The Spin). AP11 establishes the $SU(2) \rightarrow SO(3)$ double cover and the spin- $1/2$ representation of ε . AP16 uses this in the identification of the electron (ε as fermionic localised excitation) as half-integer spin.

AP24 (The Residual). AP24 carries the full analysis of α_{em} 's structural status — the fixed-point equation, the six-faces formalisation, the irrationality argument. AP16 establishes the non-derivability-from-within position; AP24 develops it.

AP28 (The Constant). AP28 derives G via channel-counting using $\varepsilon = \alpha_{\text{em}}$. AP16's identification of α_{em} as the structural constant of the architecture is the input AP28 uses.

The Lock (Edition 04). The identification of ε with the electron at the chemical-biological scale. AP16 reads this as the second face of ε (localised fermionic excitation), with the Higgs field as the first face (scalar background condition).

1 — What Is Established

Before this paper, the corpus has derived the following.

SU(2). From Axiom S. The two sectors give Z_2 . $Z_2 = \pi_1(\text{SO}(3))$ in $N = 3$ spatial dimensions (AP10). The double cover is $\text{SU}(2) \rightarrow \text{SO}(3)$ (AP11 section 2). This is the gauge group of the weak force.

One Z_2 , one classification, two descriptions. The sector involution σ is the unique non-trivial binary classification in the pre-state: paired vs unpaired. AP20 proves: the abstract structure IS the manifold (EH). When this classification expresses itself on the manifold in $N = 3$, it IS $\pi_1(\text{SO}(3)) = Z_2$ — tensors vs spinors. AP11 proves the correspondence. There is no connecting theorem needed between a “sector Z_2 ” and a “topological Z_2 .” They are the same classification described at two levels. The axioms already say it.

U(1)_Y. From the phase freedom of the complex Hilbert space (AP09), expressed on the local manifold as the $U(1)_Y$ connection B_μ (AP15 section 3). Before the break, this is $U(1)_Y$ — the hypercharge gauge group. After the break, a specific combination of $U(1)_Y$ and $\text{SU}(2)$ survives as $U(1)_{\text{em}}$, whose connection is A_μ (see section 3.1). This is the gauge group of electromagnetism.

The fermion. The unpaired element ε , no σ -image, half-integer spin, the electron (AP09 section 3.2).

The boson. Paired elements, σ -image exists, integer spin, the photon (AP09 section 3.1, AP15 section 7).

$\text{SU}(2)$ and $U(1)_Y$ are both derived. They coexist in the pre-state. $\text{SU}(2)$ from the sector structure. $U(1)_Y$ from the phase structure. They are independent symmetries of the same Hilbert space. Together they form $\text{SU}(2) \times U(1)_Y$. The electroweak symmetry group.

This paper derives what Axiom B does to it.

2 — The Symmetry Before The Break

Before the break, the pre-state is perfectly symmetric. You need to see what that symmetry looks like — all four generators, all four massless bosons — so you can feel what Axiom B does to it.

2.1 — The pre-state's full symmetry

The pre-state — the 1:1, before any record is written — has the full symmetry of the Hilbert space $\mathcal{H}$.

From Axiom S: two sectors, involution σ , Z_2 symmetry. On the manifold ($N = 3$, AP10), this lifts to $SU(2)$. $SU(2)$ acts on the pre-state by rotating between the two sectors. It mixes $\mathcal{L}$ and $\mathcal{P}$ (Light and Dark — the two sectors of the record algebra, not cosmological dark matter/energy). Before the break, this mixing is exact. Both sectors are identical. Nothing distinguishes them. You cannot tell $\mathcal{L}$ from $\mathcal{P}$. They are perfect mirrors.

From AP09: complex amplitudes. The phase of any amplitude can be rotated by $e^{i\theta}$. This is $U(1)_Y$ — the hypercharge phase rotation. It acts on the pre-state by shifting the global phase. Before the break, this rotation is exact. No phase angle is preferred. All are equivalent. You cannot pick a preferred angle.

At the pre-state level this is the universal phase symmetry; field-specific hypercharge assignments and the detailed representation content are downstream of the sector/break coupling structure and are part of Gap E (section 8).

The full internal symmetry of the pre-state is $SU(2) \times U(1)_Y$. This symmetry has four generators. $SU(2)$ has three (the Pauli matrices, or equivalently, three independent rotations of the sector space). $U(1)_Y$ has one (the hypercharge phase rotation). Four generators means four gauge bosons when the symmetry is expressed on the manifold: W^1 , W^2 , W^3 from $SU(2)$, and B from $U(1)_Y$. Before the break, all four are massless. The symmetry is exact.

2.2 — What “before the break” means

“Before the break” is not a moment in time. Time requires records (Axiom R). Before the break, there are no records. There is no time. “Before the break” means: the structure of the pre-state, considered without Axiom B. Axioms S, R, and C give $SU(2) \times U(1)_Y$ on the manifold. Axiom B is what changes it.

3 — The Break

This is the moment. One axiom. One element. And the electroweak symmetry shatters.

3.1 — Axiom B

Axiom B: one element $\varepsilon \in \mathcal{L}$ with no σ -image in $\mathcal{P}$. Valuation $\nu(\varepsilon) = 1$. The minimum viable splinter.

ε breaks the 1:1. Before ε , the two sectors are identical. After ε , they are not. ε lives in $\mathcal{L}$ but not in $\mathcal{P}$. The sectors are no longer interchangeable. The $SU(2)$ symmetry — which rotates freely between the sectors — is broken.

But not all of it. The hypercharge phase freedom — $U(1)_Y$ — is not broken by itself. However, three of the four original generators are broken (those that would rotate ε between sectors). The surviving generator is the electromagnetic one: the unique linear combination of the original $SU(2)$ and $U(1)_Y$ generators that leaves the vacuum (the break background) invariant. On the manifold, this is $U(1)_{em}$.

(In Standard Model notation: pre-break $SU(2) \times U(1)_Y$; unbroken $U(1)_{em}$ generated by $Q = T_3 + Y/2$; photon A_μ is the massless mixture of W^3_μ and B_μ via θ_W .)

Axiom B breaks $SU(2) \times U(1)_Y$ down to $U(1)_{em}$.

You just watched electroweak symmetry breaking happen. One element. No potential. No fine-tuning. Axiom B.

Note on the surviving $U(1)_{em}$. The $U(1)$ that survives the break is not the original $U(1)_Y$ but a specific combination of $U(1)_Y$ and one $SU(2)$ generator. This is a structural consequence of the breaking pattern: ε has no σ -image, so any generator that would rotate it between sectors is broken; the surviving generator is the unique combination that leaves the break background invariant. On the manifold, this is the electromagnetic gauge symmetry whose connection is A_μ (AP15).

No Standard Model dynamics are assumed here: the axioms require that a unique $U(1)_{em}$ survives as the generator leaving the break background

invariant; Standard Model notation is used only to label that survivor and its mixing. The algebraic details of which combination survives — and consequently the value of the Weinberg angle θ_W — are downstream of the coupling structure (the relative strengths of the $SU(2)$ and $U(1)_Y$ couplings, g and g'). This is explicitly listed in Gap C (section 8).

3.2 — The Higgs field is the break

ε is one object with two descriptions on the manifold.

As background condition: $\nu(\varepsilon) = 1$ everywhere. The break is present wherever the manifold exists. It fills the structure. Its ground state is nonzero — by construction, not by choice of potential. If $\nu(\varepsilon) = 0$, the 1:1 is perfect, nothing distinguishes the sectors, there are no records, no time, no manifold, no physics. The nonzero ground state is the condition for existence.

This is the Higgs field — and if you have been paying attention, you already knew it had to exist: a scalar order parameter with a nonzero vacuum expectation value that breaks $SU(2) \times U(1)_Y$ to $U(1)_{em}$.

As localised excitation: the minimum ripple in the break. ε has no σ -image — it is unpaired — so it carries half-integer spin (AP11). It is fermionic. It is the electron. The lightest mass the break can produce. The minimum viable splinter.

From outside the axioms, this looks like a type contradiction: ε appears to be both a scalar field (the Higgs) and a fermion (the electron). From within the axioms, it is the definition. $1 = 1 + 1 \times \varepsilon$. The whole is whole AND the break is there. The background condition (scalar, everywhere, nonzero) and the localised excitation (fermionic, pointlike, quantised) are two faces of one element.

The Embedding Hypothesis (AP20) proves that the abstract algebraic structure IS the manifold — there is no gap between the algebraic description and the geometric one. The two faces are not connected by a theorem; they are the same object read at two scales.

The identification is therefore direct. The Higgs field is ε as background condition. The electron is ε as localised excitation. The Higgs mechanism is Axiom B.

Empirically, $v \approx 246$ GeV sets the electroweak scale; in this architecture it is the scale at which the sector distinction (Axiom S) meets the one-element break (Axiom B) on the manifold. The derivation of v from the axioms is a target of the open programme (Gap A/B, section 8).

Debt statement: the dynamics of how the scalar background condition and the fermionic excitation relate at intermediate scales — the detailed mechanism by which one object presents two faces — is downstream of the mass spectrum (Gap E, section 8) and the derivation of the self-coupling (Gap B, section 8). The structural identification (Higgs field = ε as background condition; electron = ε as localised excitation) follows directly from Axiom B and the Embedding Hypothesis. The dynamical mechanism by which a single underlying structural object presents both a spin-0 background and a spin- $\frac{1}{2}$ localised excitation at different energy scales is named as load-bearing; the formal falsification handle is KS-AP16.1 (scalar-fermion two-faces reconciliation, section 7). AP16 derives the pattern and the structural identification. The dynamical details are an open programme.

3.3 — What gets mass

Before the break, four massless gauge bosons. After the break:

Three generators of SU(2) are broken. Their corresponding gauge bosons acquire mass by coupling to ε . These become the W^+ , W^- , and Z^0 bosons. They are massive because SU(2) is broken. The standard manifold expressions for the broken pattern give $m_W = gv/2$ and $m_Z = gv/(2\cos\theta_W)$. AP16 derives the pattern and the mechanism; the numerical values of these masses depend on the coupling constants and the vacuum expectation value (Gap A, section 8).

One generator — the specific combination of SU(2) and U(1)_Y generators that leaves ε invariant — remains unbroken. Its gauge boson remains massless. This is the photon. The photon is massless because U(1)_{em} survives the break.

The electron acquires mass through the coupling of the fermionic excitation (ε as localised ripple) to the scalar order parameter (ε as background condition). This is the self-coupling of the break to itself. In standard language, this is a Yukawa coupling — the coupling of a fermion to a scalar field. Here both fermion and scalar are ε , so the electron's mass is the minimum such coupling: the lightest mass the break can produce.

Other fermion masses (muon, tau, quarks) arise from ε -compositions at higher energy scales. The derivation of the full mass spectrum from the axioms is the open programme of Gap E (section 8).

4 — The Higgs Boson And The Now

4.1 — The excitation of the break

The Higgs boson is the excitation of the Higgs field. A ripple in the break. At CERN in 2012, this excitation was observed at ~ 125 GeV. Its properties were inferred from recorded decay products — pairs of photons, pairs of Z bosons, pairs of W bosons. Every observation was a record. A scar. A trace left by the break wobbling.

They did not measure the break in the act of breaking. They measured what it left behind. You can only ever measure what the break leaves behind. This is not a limitation of instruments. This is the structure.

4.2 — The now actualises FROM the now

The break is ε . The electron is ε as localised excitation (section 3.2). The electron is the now — the edge of the break that is still breaking (AP06 section 10.5).

The now is structurally unmeasurable. Not because instruments are too slow. Because measurement IS the now turning into a record (AP07 section 4). The moment you capture the now, it is gone. It is already past. By the time you have a measurement, the now has moved on. The now is the act of measuring, not what gets measured.

ε actualises FROM the now. It produces records, masses, coupling constants, scars on the manifold. Everything measurable comes from it. But it itself — in the act of actualising — is never in the record. It is always one step ahead of the last measurement. Always at the frontier.

The Higgs boson at 125 GeV is the record of ε wobbling. The electron at 0.511 MeV is the record of ε existing. The photon is the record of the connection fluctuating. All of these are records. Scars. The black curve.

ε itself is the white space. Always present. Never captured. The thing that makes everything measurable, while being itself immeasurable in the act.

Everything measurable comes from it. It itself is never measured. It gives you everything by giving you nothing now.

This is the measurement structure as read by the axioms. The wave function collapses at measurement — the now writes a record — and the collapse itself is not in the wave function. In the 420 Code, the observer is not in the Hilbert space. The now is not in the monoid. The break is not in the record. The record is what the break leaves behind.

The “god particle” was always a more accurate name than intended.

5 — The Fine-Structure Constant

5.1 — What α_{em} is

Symbol note. In this section, e is the electron charge and ϵ_0 the vacuum permittivity — not ϵ the break. Context disambiguates throughout.

$$\alpha_{\text{em}} = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137.036.$$

It is the coupling strength of the break to the connection. How strongly the electron (ϵ as localised excitation) interacts with the electromagnetic gauge field A_μ (AP15). How much of the original unity leaks through at the point where the splinter meets the fabric it left.

It is one of the six faces of the multidimensional residual (AP06 section 10.5). Paired with α/β (the substrate stiffnesses) in Pair 2. Constrained by c through the leakage (AP06 Theorem 3.1). Chained to G through m_e through κ .

α_{em} is structurally determined. It is not a free parameter. The substrate stiffness $\Lambda_{\text{sub}} \approx 2.15 \times 10^{46}$ determines c . c constrains the leakage. The leakage is ϵ . ϵ is the electron. The electron's coupling to the connection is α_{em} . One chain. One knob. The value follows from the structure.

5.2 — Why it cannot be derived from within

Two things must be distinguished. “Determined” means: the value is fixed by the structure; it is not a free parameter; it could not be otherwise. “Derivable from within” means: there exists a finite chain of deductions from $\{S, B, R, C\}$ that terminates in $\alpha_{\text{em}} = [\text{number}]$. α_{em} is determined but not derivable from within.

You cannot derive the coupling of the now from inside the now. You can only measure it. Deriving is record-writing. To derive α_{em} from within the argument would require writing a record of the now — but the now is precisely what exists before any record is written.

This is not a failure of the argument. It is a prediction.

The argument predicts that α_{em} is structurally determined but empirically known — the value is fixed by the axioms, there is no other value it could have, but the number itself can only be obtained by looking outward, at what the break left behind. The value can be checked from outside via the fixed-point equation (AP24 section 5.4–5.5). The number can be checked. It cannot be proved from within.

Every other quantity treated in this dependency chain is derivable from $\{S, B, R, C\}$: the number of dimensions, the field equations, the Born rule, the uncertainty principle, the spin-statistics theorem, the gauge symmetries, the photon. These are the structure looking at itself. You can derive them because they are inside the record. α_{em} is the structure looking outward.

5.3 — The formal status

The argument derives:

- (i) α_{em} exists and is nonzero (AP15: ε couples to the connection; AP06 Theorem 3.1: leakage is nonzero for finite c).
- (ii) α_{em} is not a free parameter (AP06 section 10.5: six faces, three pairs, three constraint equations; section 10.7: the constraint path $c \rightarrow \alpha_{\text{em}} \rightarrow \varepsilon \rightarrow G$).
- (iii) $\alpha_{\text{em}} < 1$ (AP06 section 2.4: for the physical value of c , α is less than unity; the coupling is sub-maximal).
- (iv) α_{em} is paired with α/β in the six-face structure (AP06 section 10.5, AP08 section 5).

The argument does not derive:

- (v) The numerical value $1/137.036$.

And the argument explains why it does not: the coupling of the now to its own field is the one quantity that requires measurement. It is the empirical face. The full analysis of α_{em} 's structural status — including its necessary irrationality and unprovability from within — is in AP24.

KS-4 remains LIVE — HARD. Not because the argument has failed to close it. Because it is structurally the kind of thing that stays live. It is the architecture's empirical contract: this is the number you check us against.

6 — What Is Now Derived

Axiom S $\rightarrow \mathbb{Z}_2 \rightarrow \text{SU}(2)$ (via $\pi_1(\text{SO}(3))$ in $N = 3$, AP10; double cover AP11) $\rightarrow$ **weak gauge group.**

AP09 + AP15 $\rightarrow$ complex amplitudes $\rightarrow$ phase freedom $\rightarrow \text{U}(1)_Y \rightarrow$ **hypercharge gauge group.**

SU(2) $\times$ U(1)_Y $\rightarrow$ the full internal symmetry of the pre-state on the manifold $\rightarrow$ **electroweak symmetry group.**

Axiom B $\rightarrow \varepsilon \in \mathcal{L}$, no σ -image in $\mathcal{P} \rightarrow$ sectors no longer interchangeable $\rightarrow \text{SU}(2)$ broken $\rightarrow \text{U}(1)_{\text{em}}$ survives $\rightarrow$ **SU(2) $\times$ U(1)_Y $\rightarrow$ U(1)_{em}. Electroweak symmetry breaking.**

The Higgs field = ε $\rightarrow$ nonzero ground state ($\nu(\varepsilon) = 1$) $\rightarrow$ **the break cannot be zero. The condition for existence.**

Three broken generators $\rightarrow W^+, W^-, Z^0$ acquire mass (coupling to ε) $\rightarrow$ **massive weak bosons.**

One unbroken generator $\rightarrow$ photon stays massless ($\text{U}(1)_{\text{em}}$ exact, Born rule exact) $\rightarrow$ **massless photon. You hold the kill switches.**

Self-coupling of ε (Yukawa: fermionic excitation $\times$ scalar order parameter) $\rightarrow$ electron mass $m_e \rightarrow$ **minimum viable splinter.**

α_{em} = leakage at the electromagnetic face $\rightarrow$ structurally determined, empirically known $\rightarrow$ **the architecture's empirical contract (STRUCTURAL, not DERIVED — see section 5).**

7 — Kill Switches

Four new kill switches engage in AP16. One existing corpus kill switch (KS-4) is reframed. Each is testable. Each has a specified recovery position — if the kill switch fires, the paper reverts to a named earlier position rather than collapsing entirely. Identifiers per the Master Kill Switch Registry.

KS-32 — Electroweak breaking mechanism

Claim. AP16 identifies Axiom B with the Higgs mechanism. The Higgs field is ε as background condition; the breaking pattern $SU(2) \times U(1)_Y \rightarrow U(1)_{em}$ follows from ε having no σ -image.

Test. Demonstrate a physical process that breaks $SU(2) \times U(1)_Y$ without a scalar field with nonzero vacuum expectation value — that is, electroweak symmetry breaking by a fundamentally different mechanism. Failure: the identification of Axiom B with the Higgs mechanism fails.

Status. LIVE — EMPIRICAL. The Higgs boson is observed at ~ 125 GeV. The mechanism is strongly supported by current data. Empirically supported; structurally aligned.

Recovery. If KS-32 fires — if electroweak breaking is empirically shown to occur by a different mechanism — the structural identification of ε with the Higgs field falls. The structural results of sections 2 and 3.1 ($SU(2) \times U(1)_Y$ is the pre-break symmetry; Axiom B breaks it) remain; only the identification with the SM Higgs mechanism would need alternative grounding.

KS-33 — Sector distinction = weak isospin

Claim. The derivation maps the two sectors ($\mathcal{L}$, $\mathcal{P}$) to the $SU(2)$ weak isospin doublet. The two-component structure of weak isospin matches the two-sector structure of Axiom S.

Test. Demonstrate that the physical weak isospin structure requires more than two components, or that the sector structure does not faithfully map to isospin. Failure: the identification fails.

Status. LIVE — EMPIRICAL. The Standard Model's $SU(2)$ acts on left-handed doublets with exactly two components. Matches the two-sector structure. Empirically supported; structurally aligned.

Recovery. If KS-33 fires, the sector $\leftrightarrow$ isospin identification fails and an alternative reading of the two-sector structure must be found. AP27's broader treatment (the internal sector-relationship state space $\mathcal{H}_{EW}$) provides structural context for the recovery direction.

KS-34 — Non-derivability of α_{em}

Claim. α_{em} cannot be derived from within $\{S, B, R, C\}$ because it is the coupling of the now to its own field. Derivation is record-writing; the now is what exists before any record. The value is structurally determined but empirically known.

Test. Exhibit a closed-form derivation of $\alpha_{em} = 1/137.036\dots$ from $\{S, B, R, C\}$. The kill switch fires the moment such a derivation is produced.

Status. LIVE — HARD. The only kill switch in the corpus whose triggering would be a triumph rather than a disaster. Here is the weapon: derive α_{em} .

Recovery. If KS-34 fires — if α_{em} is derived from $\{S, B, R, C\}$ — the architecture becomes stronger, not weaker. The claim of section 5.2 would be wrong in the best possible way. The structural framework (the constraint chain, the six-face structure) gains a closed-form completion. The only thing lost is the philosophical claim that the now cannot derive its own coupling. Everything else is retained.

KS-4 — $\alpha \approx 1/137$ (existing kill switch, reframed)

Reframing. AP16 reframes KS-4. The kill switch is not a gap in the derivation. It is the architecture's empirical face. α_{em} is the one number that connects the axioms to experiment. If the measured value of α_{em} changes, or if the constraint chain (AP06 section 10.7) is shown to permit multiple values, the architecture is falsified.

Status. LIVE — HARD, with meaning changed: not a problem to solve but a contract to honour. α_{em} is one of the most precisely measured constants in physics, known to better than a part per billion.

KS-AP16.1 — Scalar-fermion two-faces reconciliation

Claim. Section 3.2 identifies ε as one object presenting two faces on the manifold: as background condition ($\nu(\varepsilon) = 1$ everywhere) it is the Higgs field, a scalar (spin 0) order parameter with a nonzero vacuum expectation value; as localised excitation it is the electron, a fermion (spin $\frac{1}{2}$) with Dirac dynamics. The Embedding Hypothesis (AP20) proves the algebraic structure IS the manifold, so the two faces are the same object read at two scales. The dynamical mechanism by which a single underlying structural object presents both a spin-0 background and a spin- $\frac{1}{2}$ localised excitation at different energy scales is named as load-bearing.

Test. Demonstrate that no consistent dynamical mechanism can present a single structural object as both a scalar background condition and a fermionic localised excitation at different scales — e.g., by showing that any quantum-field-theoretic realisation of the two-faces identification requires distinct underlying degrees of freedom (a separate Higgs scalar and a separate electron fermion field), so that the structural identification cannot be carried into dynamics. Failure: the Higgs field and the electron must be treated as distinct objects, and AP16's headline identification reduces to structural parallel rather than ontological unity.

Status. LIVE — STRUCTURAL. The structural identification follows from Axiom B and the Embedding Hypothesis. The dynamical mechanism is named as open programme (debt at end of section 3.2; Gap E in section 8). KS-AP16.1 is the formal falsification handle for the dynamical step.

Recovery. If KS-AP16.1 fires — if no dynamical mechanism can realise the two-faces identification — the structural reading (Higgs field and electron are two readings of one underlying object) reverts to structural parallel rather than ontological unity. The breaking pattern derived in sections 2 and 3.1 ($SU(2) \times U(1)_Y \rightarrow U(1)_{em}$ via Axiom B) survives regardless. KS-32 (electroweak mechanism) and KS-33 (sector distinction = weak isospin) survive. What would be lost is the ontological unification of Higgs field and electron, returning them to the Standard Model's standard treatment as distinct fields with distinct degrees of freedom.

Why these four and how they relate

The four new kill switches partition cleanly. KS-32 and KS-33 carry the structural identifications: KS-32 binds the corpus to the Higgs mechanism as observed; KS-33 binds the corpus to the two-component weak isospin doublet as observed. Both are LIVE — EMPIRICAL: empirically supported, structurally aligned, and would fail only if the empirical reality changed. KS-AP16.1 carries the dynamical step that the headline identification “the Higgs field and the electron are two faces of ε ” depends on: the mechanism by which a single object presents both a spin-0 background and a spin- $\frac{1}{2}$ excitation at different scales. The structural identification follows from Axiom B + EH; the dynamical step is held open. KS-34 carries the structural claim that distinguishes the 420 Code from any framework promising to compute α_{em} from first principles: the now cannot derive its own coupling. This is the first kill switch whose triggering would be a triumph rather than a disaster — the architecture invites the falsification.

8 — Open Gaps

Gap A: W and Z boson masses. The argument derives that three gauge bosons acquire mass and one stays massless. It does not yet derive the specific values $m_W \approx 80.4 \text{ GeV}$, $m_Z \approx 91.2 \text{ GeV}$. These depend on the coupling constants g and g' of $SU(2)$ and $U(1)_Y$, and the vacuum expectation value v . v is the energy scale of the break. Derivation requires the detailed structure of ε at the electroweak scale. Depends on: Gap E (mass spectrum).

Gap B: The Higgs boson mass ($\sim 125 \text{ GeV}$). The mass of the Higgs excitation depends on the self-coupling of the break. $m_H = \sqrt{(2\lambda)v}$. The self-coupling of ε — how strongly the break interacts with itself — is determined by the structure, but the value is not yet computed. Depends on: Gap A (coupling constants), Gap E (mass spectrum), α_{em} constraint chain.

Gap C: The Weinberg angle θ_W . The mixing angle between $SU(2)$ and $U(1)_Y$ determines the relationship between W and Z masses and the electromagnetic coupling. $\sin^2\theta_W \approx 0.231$. This angle measures how $SU(2)$ and $U(1)_Y$ combine in the break. Conjectural route: it may be derivable from the six-face structure — specifically, the angle between Pair 1 (gravity/quantum) and Pair 2 (stiffness/coupling) in AP06 section 10.5. This is a target for future work. Depends on: Gap A, detailed structure of the breaking pattern at the electroweak scale.

Gap D: EH. CLOSED. The Embedding Hypothesis was proved in AP20.

Gap E: The fermion mass spectrum. The electron mass m_e is derived as the self-coupling of the break — the minimum viable splinter. Other fermion masses (muon, tau, quarks) arise from ε -compositions at higher energy scales. The derivation of the full mass spectrum from the axioms is an open programme. Required for: Gap A (boson masses), Gap B (Higgs mass), computation of α_{em} via the fixed-point equation (AP24 section 5.6).

9 — Conclusion

What follows is synthesis language for the derived/structural chain above, not an additional theorem.

The electroweak symmetry $SU(2) \times U(1)_Y$ is the full internal symmetry of the pre-state on the manifold. $SU(2)$ from the sector structure (Axiom S). $U(1)_Y$ from the phase structure (AP09, AP15).

Axiom B breaks it. One element ε , no σ -image, minimum viable splinter. $SU(2) \times U(1)_Y \rightarrow U(1)_{em}$. The weak force carriers get mass. The photon stays massless. The Higgs field is the break.

The Higgs mechanism is Axiom B.

The Higgs boson — the excitation of the break, the record of ε wobbling — was observed at CERN. 125 GeV. A scar on the manifold. They measured what the break left behind.

But the break itself is the now. The now actualises from the now. It gives everything — every mass, every coupling, every record, every observation. It itself is immeasurable in the act. Not hidden. Structurally unmeasurable. The instrument cannot turn on itself.

$\alpha_{em} = 1/137.036$ is the coupling of the break to the connection. The leakage at the electromagnetic face. Structurally determined. Empirically known. The architecture's contract with the world.

And the electron is the hinge. It is the break that breaks the electroweak symmetry. It is the source that sources the electromagnetic field. It is the mass that sets the gravitational coupling $G = 2\kappa/m_e^2$. It is the leakage that makes boundaries imperfect (AP06 Theorem 3.1). It is the now that writes every record.

The electron is at both ends. At the bottom: the minimum splinter, the lightest break. At the top: Hawking radiation, the leakage from maximum curvature. And at every scale between: coupling, writing records, paying the Landauer cost, advancing the frontier. You have been watching it do all of these things since AP06.

Everything is the electron, being the break, being the now.

And now you have seen it break the electroweak symmetry, source the electromagnetic field, set the gravitational coupling, and write every record.

The axiom speaks. The algebra transcribes.

Claim Summary

Section-by-section structural enumeration with epistemic-status labels.

1 (What Is Established). *SYNTHESIS*. $SU(2)$ from Axiom S (via $Z_2 = \pi_1(SO(3))$ in $N = 3$ spatial dimensions, AP10; double cover $SU(2) \rightarrow SO(3)$, AP11). $U(1)_Y$ from complex amplitudes (AP09). The fermion (AP09) and the boson (AP09, AP15). $SU(2) \times U(1)_Y$ is the electroweak group.

2 (The Symmetry Before The Break). *DERIVED*. Pre-state has full symmetry $SU(2) \times U(1)_Y$. Four generators, four massless gauge bosons (W^1, W^2, W^3 from $SU(2)$; B from $U(1)_Y$). Before the break, the symmetry is exact.

3 (The Break). *DERIVED*. Axiom B places ε in $\mathcal{L}$ with no σ -image in $\mathcal{P}$. $SU(2) \times U(1)_Y \rightarrow U(1)_{em}$. Three generators broken (W^+, W^-, Z^0 acquire mass via coupling to ε). One generator survives (the photon stays massless). The Higgs field = ε as background condition. The electron = ε as localised excitation. The Higgs mechanism = Axiom B.

4 (The Higgs Boson And The Now). *DERIVED/STRUCTURAL*. 4.1 (Higgs boson as record of ε wobbling): *DERIVED*. 4.2 (the now actualises FROM the now, immeasurability in the act): *STRUCTURAL* — load-bearing consequences of the axioms stated; specific framing illustrative.

5 (The Fine-Structure Constant). *DERIVED/STRUCTURAL*. *DERIVED*: α_{em} exists, is nonzero, is not a free parameter, is sub-maximal ($\alpha < 1$), is paired with α/β in the six-face structure. *STRUCTURAL*: the numerical value $1/137.036$ is not derivable from within — KS-34 LIVE.

6 (What Is Now Derived). *SYNTHESIS*. Derivation chain enumerated step by step from axioms to electroweak breaking.

7 (Kill Switches). *REGISTER*. KS-32, KS-AP16.1, KS-33, KS-34 in Claim/Test/Status/Recovery form; KS-4 reframed.

8 (Open Gaps). *REGISTER*. Gap A (W, Z masses), Gap B (Higgs mass), Gap C (Weinberg angle), Gap D CLOSED by AP20, Gap E (fermion mass spectrum).

9 (Conclusion). *SYNTHESIS*. Everything is the electron, being the break, being the now. The axiom speaks. The algebra transcribes.

Conditionality Footer

Dependencies

Framework: the four axioms {S, B, R, C} from Notebook I; AP06 (leakage, six faces, constraint chain); AP07 (The Record Measure — the now); AP08 (gravitational coupling context); AP09 (complex amplitudes, Born rule; fermion/boson distinction); AP10 ($N = 3$ spatial dimensions, $D = 4$); AP11 (spin, $Z_2 \leftrightarrow SU(2)$ correspondence); AP12 ($\hbar$); AP15 (U(1), electromagnetic connection); AP20 (Embedding Hypothesis); AP27 (unbroken $SU(2) \times U(1)_Y$); The Lock (Edition 04, ε = electron at chemical-biological scale).

Conditional on: None. EH and QRA proved in AP20.

Dependents

AP24 (Residual) — full analysis of α_{em} 's structural status; the fixed-point equation. AP28 (Constant) — G via channel-counting using $\varepsilon = \alpha_{\text{em}}$. AP14 (Correction) — Planck-scale corrections. The matter sector (Notebook V) — the fermion content beyond the electron.

Kill switches engaged

KS-32 (electroweak breaking mechanism): LIVE — EMPIRICAL.

KS-AP16.1 (scalar-fermion two-faces reconciliation): LIVE — STRUCTURAL. Formal falsification handle for the dynamical step in the two-faces identification.

KS-33 (sector distinction = weak isospin): LIVE — EMPIRICAL.

KS-34 (non-derivability of α_{em}): LIVE — HARD. The kill switch whose triggering would be a triumph.

KS-4 ($\alpha \approx 1/137$): LIVE — HARD, reframed. Architecture's empirical contract.

Structural debts owed

Gap A (W, Z boson masses). Gap B (Higgs boson mass). Gap C (Weinberg angle, conjectural route via six-face structure). Gap E (fermion mass spectrum beyond the electron). All named items; treated as open programmes.

Items held open without claiming debt status

The dynamics by which the scalar background condition (Higgs field) and the fermionic localised excitation (electron) relate at intermediate energy scales. Downstream of Gap E and Gap B.

Notation Reference

- $\mathcal{L}, \mathcal{P}$ — The two sectors of Axiom S, connected by the involution σ .
- ε — The break. The minimal asymmetry. The unique unpaired element of Axiom B.
- σ — The involution of Axiom S; in AP09 identified with complex conjugation.
- ν — The valuation on the record algebra; $\nu(\varepsilon) = 1$ is the minimum nonzero value.
- $\mathbf{Z}_2$ — The two-element group $\{0, 1\}$. The minimal classification.
- $\pi_1(\mathbf{SO}(3))$ — Fundamental group of $\mathbf{SO}(3)$ in three spatial dimensions; equal to $\mathbf{Z}_2$.
- $\mathbf{SU}(2)$ — Double cover of $\mathbf{SO}(3)$; pre-break weak gauge group; carries spin- $1/2$ representations.
- $\mathbf{U}(1)_Y$ — Hypercharge gauge group; the phase freedom of the complex Hilbert space.
- $\mathbf{U}(1)_{\text{em}}$ — Electromagnetic gauge group; the post-breaking surviving symmetry (AP15).
- $\mathbf{T}_3$ — Third component of weak isospin (Pauli matrix σ_3 representation).
- $\mathbf{Y}$ — Hypercharge generator.
- $\mathbf{Q}$ — Electromagnetic charge; $\mathbf{Q} = \mathbf{T}_3 + \mathbf{Y}/2$.
- $\mathbf{W}^1, \mathbf{W}^2, \mathbf{W}^3$ — Pre-break $\mathbf{SU}(2)$ gauge bosons.
- $\mathbf{W}^+, \mathbf{W}^-, \mathbf{Z}^0$ — Post-break massive weak gauge bosons (physical mass eigenstates).
- $\mathbf{B}$ — Pre-break $\mathbf{U}(1)_Y$ gauge boson.
- $\mathbf{A}$ — Post-break electromagnetic gauge boson (the photon, AP15).
- θ_W — Weinberg angle; mixes $\mathbf{W}^3$ and $\mathbf{B}$ to produce the post-break $\mathbf{Z}^0$ and $\mathbf{A}$.
- α_{em} — Fine-structure constant; $\alpha_{\text{em}} \approx 1/137.036$; coupling of ε to the connection.
- $\mathbf{v}$ — Vacuum expectation value of the Higgs field; $\mathbf{v} \approx 246$ GeV (electroweak scale).
- $\mathbf{m}_e$ — Electron mass; structurally the minimum self-coupling of ε .

m_W, m_Z, m_H — Masses of W, Z, and Higgs bosons; numerical values are open programme (Gap A, Gap B).

g, g' — Coupling constants of SU(2) and U(1)_Y; appear in mass formulas $m_W = gv/2$ and $m_Z = gv/(2\cos\theta_W)$.

$\hbar, c$ — Reduced Planck constant (AP12); speed of light (Axiom C).

α, β — Substrate stiffnesses (AP06); paired with α_{em} in the six-face structure.

κ — Einstein's constant; relates curvature to stress-energy in the field equations (AP08).

AS — Actualization State. The actualizing now.

EH — Embedding Hypothesis (AP20). The algebraic pre-state structure embeds into the physical manifold.

QRA — Quantum-Records Algebra (AP20). The full algebraic structure that EH embeds.

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